

LAB REPORT

[METAL 1: DEFORMATION AND STRENGTHENING]



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1. Introduction

The service performance of metallic materials largely depends on the coupling between their microstructure and mechanical properties. For aero-engines operating under high-temperature and high-stress conditions, titanium alloys and nickel-based superalloys have become key structural materials owing to their high specific strength and excellent high-temperature stability. To systematically understand the intrinsic relationship among heat-treatment processes, microstructure, and macroscopic properties, this experiment is designed around two components: metallographic microstructural observation and hardness testing of metals. The former focuses on comparing the evolution of microstructural morphology under different heat-treatment conditions, while the latter quantitatively characterizes the local resistance of the material to plastic deformation using hardness as a representative mechanical parameter.

1.1 Metallographic Examination

Metallographic examination is one of the most fundamental and important characterization methods in materials science and engineering. By observing polished and etched metallic specimens under an optical microscope, one can obtain microstructural information such as **grain size, phase constitution, precipitate distribution, and twinning**, and thereby analyze trends in the material's strength, ductility, and toughness.

In this experiment, IMI834 titanium alloy and GH4145 nickel-based superalloy, both commonly used in aero-engines, are selected as the research objects. IMI834 is an $\alpha+\beta$ dual-phase titanium alloy, in which different heat-treatment processes can produce several typical microstructural morphologies, including equiaxed structures, Widmanstätten structures, and bimodal structures, thus providing a good platform for studying the influence of heat-treatment schedules on grain morphology and phase distribution. GH4145 nickel-based superalloy, on the other hand, exhibits an equiaxed grain structure with a face-centered cubic lattice; after different heat treatments, the grain size and the number of twins change significantly, enabling quantitative analysis of the relationship between grain refinement and mechanical performance. Through systematic observation and comparison of the above specimens, this experiment aims

to master the microstructural characteristics of typical aerospace metallic materials and to understand their correlation with service performance.

1.2 Hardness Testing

Hardness is an important parameter for evaluating the resistance of metallic materials to localized plastic deformation and is often closely correlated with properties such as strength and wear resistance. In this experiment, three indentation-based hardness testing methods will be used: **Rockwell hardness**, **Brinell hardness**, and **micro-Vickers hardness**. (Figure 1.) Rockwell hardness determines the hardness value by measuring the difference in indentation depth under a minor load and a major load, and is suitable for the rapid evaluation of materials with medium to high hardness. Brinell hardness uses a steel ball or cemented carbide ball indenter, which is pressed into the material surface under a specified test load and dwell time; the average compressive stress is then calculated from the indentation diameter, making it suitable for large-volume specimens and coarse-grained materials. Micro-Vickers hardness employs a diamond pyramid indenter under a small load, enabling the characterization of hardness variations in thin layers, local regions, and individual microstructural constituents.

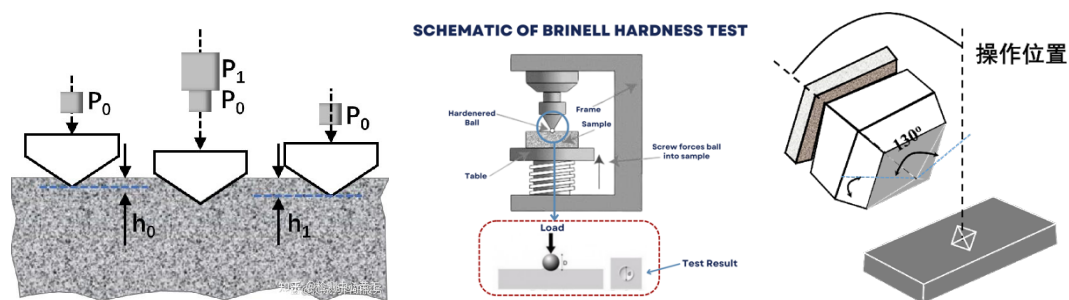


Figure 1. Schematic diagram of the Rockwell, Brinell and Vickers hardness test.

By performing multiple hardness tests on the same or similar materials and comparing the results, this experiment aims to deepen understanding of the applicable ranges, advantages and disadvantages, and conversion relationships among different hardness scales, while also training students in the proper use of hardness testers and the processing of experimental data.

2. Materials

2.1 Metallographic Examination

In this experiment, the specimens to be tested are IMI834 titanium alloy and GH4145 nickel-based superalloy, each subjected to different heat treatments to produce six samples, as shown in the table below.

Table 1. 6 samples.

Alloy	Technology for heating processing		
IMI834	Equiaxed	Widmanstätten	Bimodal
GH4145	Initial	1125°C	1150°C

The experimental equipment used in this study is an optical metallographic microscope equipped with a digital imaging system, which is employed to observe, analyse, and capture images of the prepared metallographic microstructures.

2.2 Hardness Testing

The materials and equipment used in this experiment include standard **hardness test blocks** and **hardness testers**.

A. Rockwell hardness tester

(Shanghai Yanrun Opto-Mechatronics Technology Co., Ltd., model: HR-150A)

Equipped with a diamond cone indenter and a steel ball indenter, capable of applying multiple test loads to enable rapid measurements and direct reading of hardness values on scales such as HRC and HRB.

The standard hardness test block has a hardness of 47.1 HRC (No. R893457).



Figure 2. Rockwell hardness tester and the standard hardness test block.

B. Micro-Vickers hardness tester

(Shanghai Yanrun Opto-Mechanical Technology Co., Ltd., model: HVT-1000A)

Equipped with a diamond pyramid indenter, capable of applying precise micro-loads and measuring micron-scale indentations through a high-magnification optical system, and used to determine the HV values of micro-regions.

The standard hardness test block has a hardness of 258 HV0.05.



Figure 3. Micro-Vickers hardness tester and the standard hardness test block.

3. Procedures

3.1 Metallographic Examination

(1) Powering on the instrument and checking the environment: Ensure that the optical microscope is placed stably on a flat laboratory bench, with no significant vibration or direct light interference in the surroundings. Switch on the microscope and the light source, and allow about 2–3 minutes for warm-up so that the light output becomes stable. Adjust the brightness to a medium–low level to avoid excessive illumination that could affect observation and image quality.

(2) Sample preparation: Take the polished and etched metallographic specimens (IMI834 and GH4145) from the sample box, holding them only by the edges to avoid touching the examined surface. If there is dust or fingerprints on the surface, gently wipe it with clean lens tissue to ensure that the observation area is clean and flat.

(3) Mounting and adjusting the sample: Place the specimen with the polished surface facing upwards at the center of the stage and secure it. Select a low-magnification objective lens (50×) and use the coarse focus knob to adjust the stage height until the grain contours become clearly visible through the eyepiece; then use the fine focus knob to further optimize the image. Adjust the aperture and

brightness according to the field brightness so that the grain boundaries are sharp and the contrast is appropriate. First, scan the sample at low magnification to observe the grain size and uniformity, then successively switch to medium and high magnifications (100×, 200×), and refocus with the fine focus knob after each objective change.

(4) Image acquisition and documentation: Start the microscope's imaging software on the computer, select representative fields of view with typical grain morphology and without obvious scratches or contamination, and acquire and save at least one clear micrograph. When naming the files, indicate the material name, heat treatment condition, and magnification. In the lab notebook, briefly record the imaging location and its microstructural features to facilitate subsequent analysis.

3.2 Hardness Testing

Measure the hardness of different metallic specimens or specimens subjected to different heat-treatment conditions using the Rockwell hardness tester and the micro-Vickers hardness tester, and compare the results.

3.2.1 Rockwell Hardness Testing

(1) Preparation of instrument and specimens: Rockwell hardness testing is carried out using an HR-150 hardness tester with the HRC scale (corresponding to a diamond cone indenter and a total test load of 150 kgf). Before testing, clean the surface of the specimens with lint-free wipes or a soft cloth to remove oil, dust, and loose scale, ensuring that the test area is flat, clean, and free from obvious scratches or pits.

(2) Placing and roughly leveling the specimen: Place the specimen at the center of the tester's anvil with the test surface facing upward, avoiding regions where previous indentations are present. Slowly rotate the elevating handwheel to raise the anvil until the specimen makes slight contact with the indenter. Observe the pointer carefully during contact and ensure that there is no sudden, large jump, which would indicate that the specimen is unstable or not in good contact.

(3) Applying the minor load: Move the control lever to the "minor load" position so that the instrument automatically applies the specified minor test force. Maintain this load for about 10 s until the pointer becomes essentially stable, then manually rotate the dial so that the pointer is set to zero, ensuring accurate subsequent readings.

(4) Applying the major load and holding: On the basis of a stable minor load, continue to slowly rotate the elevating handwheel. After observing approximately three revolutions of the dial, shift the control lever from the “minor load” position to the “total load” position, allowing the instrument to automatically apply the full test load of 150 kgf. Maintain this total load for about 10 s.

(5) Unloading and reading the result: After the holding time has elapsed, move the control lever back to the “minor load” position to remove the major load while keeping the minor load applied. At this point, the indentation on the specimen surface has been formed and the pointer should be stable. Read the HRC hardness value directly from the dial and record it completely in the data sheet. Finally, return the control lever to its initial position, lower the anvil appropriately using the handwheel, and remove the specimen.

3.2.2 Micro Vickers Hardness Testing

(1) Instrument setup and parameter selection: Switch on the micro-Vickers hardness tester, as well as its built-in optical system and display. Select a test load of 0.98 N and set the loading time to 10–15 s.

(2) Specimen placement and focusing: Place the specimen on the stage of the microhardness tester with the test surface facing upward. First use a low-magnification objective to locate a flat area free of scratches and pores as the test point. Then switch to an objective with appropriate magnification (40×) and use the focusing knobs to make the surface features clear with suitable contrast.

(3) Indentation loading: After confirming the test point, operate the software to start loading and switch to the indenter mode. Wait for about 20 s; once loading is completed, the indenter will automatically lift, leaving a diamond-shaped indentation.

(4) Measuring the indentation diagonals and calculating hardness: Switch back to observation mode and observe the newly formed indentation through the microscope. Use the software to frame the diamond-shaped region and manually adjust its position. The instrument will automatically calculate the hardness value HV according to the Vickers hardness formula and display it on the screen.

4. Results and observations

4.1 Metallographic Examination

A. GH4145 Sample

GH4145 (initial) exhibits a partially recrystallized equiaxed grain structure. The grains have an irregular polyhedral morphology, and local grain boundaries show serrated distortions. Typical face-centered cubic $\{111\} \langle 112 \rangle$ deformation twins can be observed inside the grains; these twins are arranged in parallel plate-like forms, and their sharp twin boundaries intersect the primary grain boundaries at an angle of approximately 60° . The number of precipitates is low; they appear as coarse agglomerates with poor spatial dispersion. The grain size ranges from 8 to 12 μm .

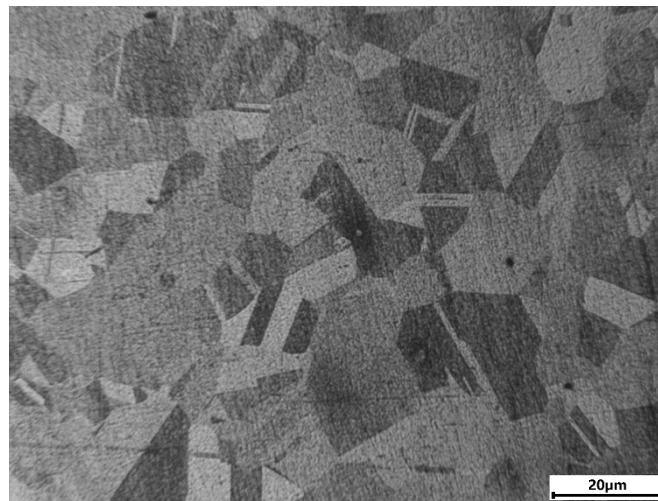


Figure 4. Metallographic microstructure of GH4145 in the initial condition.

GH4145 (1125 °C solution-treated condition) exhibits a fully recrystallized equiaxed grain structure. The grain boundaries are continuous and well-defined, but locally there are faint banded features, indicating slight compositional segregation. The number and size of intragranular precipitates lie between those of the two aforementioned specimens, and small clusters (2–3 particles) showing weak coagulation can be observed locally. No twinning is observed. The grain size ranges from 12 to 14 μm .

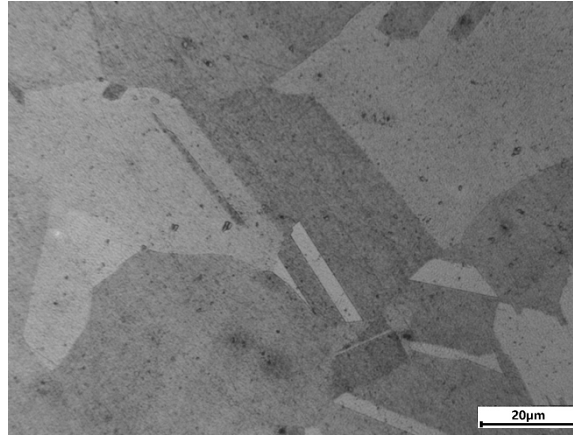


Figure 5. Metallographic microstructure of GH4145 after solution treatment at 1125 °C

GH4145 (1150 °C solution-treated condition) exhibits a fully recrystallized and homogeneous equiaxed grain structure. The grain boundaries are continuous and straight, corresponding to a low-energy, stable state. A large number of fine, granular precipitates are uniformly dispersed within the grains, with an equivalent diameter of approximately 0.5–1 μm, and no agglomeration is observed. No twin structures are detected. The grain size ranges from 10 to 15 μm.

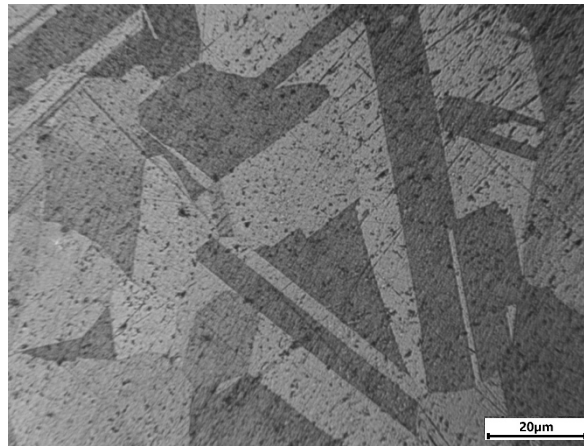


Figure 6. Schematic metallographic microstructure of GH4145 after solution treatment at 1150 °C

Grain size evolution mechanism: the grain size follows the sequence: initial < 1125 °C quenched condition < 1150 °C quenched condition. Grain growth is essentially a process of grain boundary migration, whose rate is jointly controlled by the driving force for boundary motion and atomic diffusion kinetics. As the heat-treatment temperature increases, the atomic diffusion coefficient increases exponentially, significantly enhancing grain boundary mobility and thereby accelerating grain growth.

B. IMI834

Equiaxed structure: this microstructure consists of fully recrystallized clusters of equiaxed grains. The primary α phase appears polygonal, with an equivalent circle diameter of 5–8 μm . These α_p grains, shown in light gray contrast, are dispersed within a dark-gray transformed β matrix. The phase boundaries are sharp and well-defined, with no compositional segregation or particle agglomeration. The prior β grain boundaries have completely decomposed, leaving only the fine boundaries of the equiaxed grains. No defects such as dislocations or inclusions are observed within the field of view, which is consistent with a stable microstructure formed by prolonged annealing below the β -transus temperature.

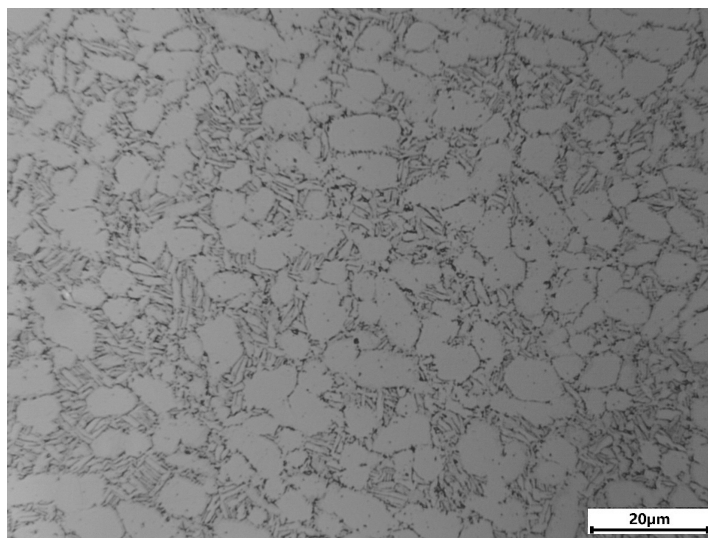


Figure 7. Schematic metallographic microstructure of the equiaxed structure in IMI834

Bimodal structure: this microstructure exhibits a mixed morphology of “primary α_p + Widmanstätten-type secondary α phase (α_s).” The α_p phase consists of equiaxed grains with an equivalent circle diameter of 4–6 μm , appearing light gray and preferentially nucleating near the prior β grain boundaries. The α_s phase appears as short, plate-like structures in dark gray, growing in random directions with blunt ends. Portions of the prior β grain boundaries are retained and appear as intermittent, faint bright lines. A narrow compositional transition zone exists at the α_p/α_s interfaces, reflecting the microstructural evolution induced by sub-transus annealing at intermediate temperatures combined with a moderate cooling rate.

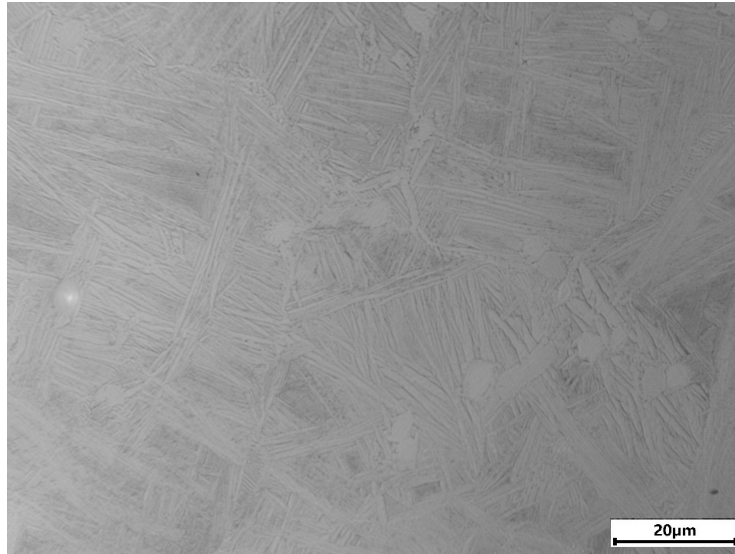


Figure 8. Metallographic microstructure of the bimodal structure in IMI834

Widmanstätten structure: this microstructure is dominated by directionally grown Widmanstätten α lath grains, with no primary α_p grains observed. The Widmanstätten α (α_v) laths appear dark gray and slender, following the Burgers orientation relationship between the α and β phases. Within the prior β grains, they form colonies of parallel laths, with misorientation angles of about 30–60° between different colonies. The prior β grain boundaries are clear and intact, serving as distinct interfaces separating colonies of different orientations. The edges of the α_v laths are straight, without branching or defects, which is characteristic of products formed by heating above the β -transus temperature followed by slow cooling.

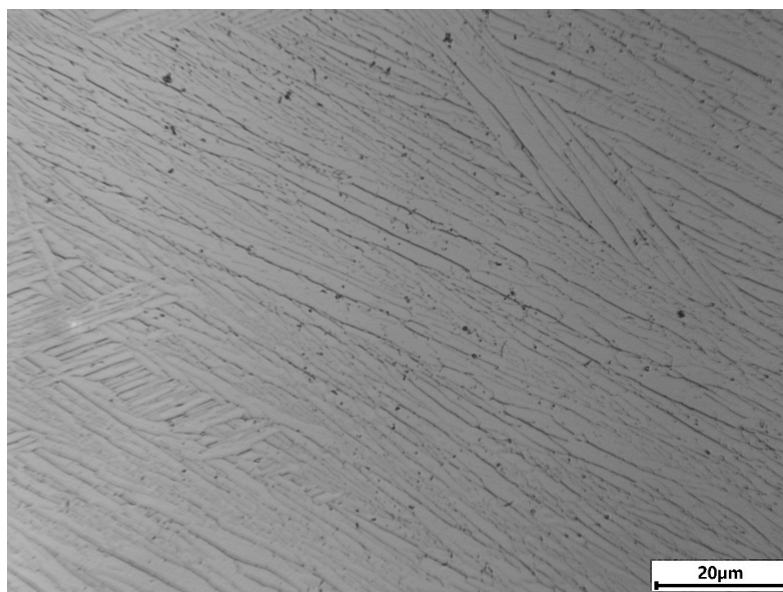


Figure 9. Metallographic microstructure of IMI834 with Widmanstätten structure

Summary of key differences:

For the IMI834 ($\alpha+\beta$) titanium alloy, the essential differences among the equiaxed, bimodal, and Widmanstätten structures lie in the morphology of the α phase, the state of the prior β grain boundaries, and the corresponding formation processes.

The α phase morphology varies significantly: from uniformly distributed polygonal primary α_p grains (equiaxed structure), to a mixed morphology of α_p and short plate-like secondary α_s (bimodal structure), and finally to a Widmanstätten structure composed solely of elongated, directionally grown α_v laths. Accordingly, the prior β grain boundaries are completely decomposed and invisible in the equiaxed structure, partially retained but blurred in the bimodal structure, and clearly preserved as grain boundaries in the Widmanstätten structure.

These differences originate directly from the heat-treatment parameters. The equiaxed structure is formed by prolonged annealing below the β -transus temperature; the bimodal structure results from sub-transus annealing followed by moderate cooling; and the Widmanstätten structure requires heating above the β -transus followed by slow cooling, which promotes directional growth of α_v laths. Therefore, the evolution of the microstructure is governed by specific combinations of temperature, holding time, and cooling rate.

4.2 Hardness Testing

A. Rockwell hardness test results and analysis



Figure 10. Schematic diagram of the Rockwell hardness test

Rockwell hardness testing was carried out using a Rockwell hardness tester. The measured value was 47.0 HRC, while the corresponding standard value was 47.1 HRC. The deviation between the two is only 0.1 HRC, indicating excellent consistency and that the result lies well within the allowable tolerance range of the test. The possible causes of this slight deviation include:

Operational details: Minor misalignment between the indenter and the specimen surface, or slight instability in the loading rate during the application of the test force.

Specimen surface condition: Even small local variations in surface roughness within the test area can affect the contact state of the indentation, leading to slight deviations in the measured hardness value.

Dial gauge reading error: Errors may arise from zero-offset of the analog dial, small inconsistencies in scale accuracy, parallax error during reading, or reading the value before the pointer has fully stabilized. Such errors typically cause fluctuations within approximately ± 0.5 hardness units.

B. Vickers hardness test results and analysis

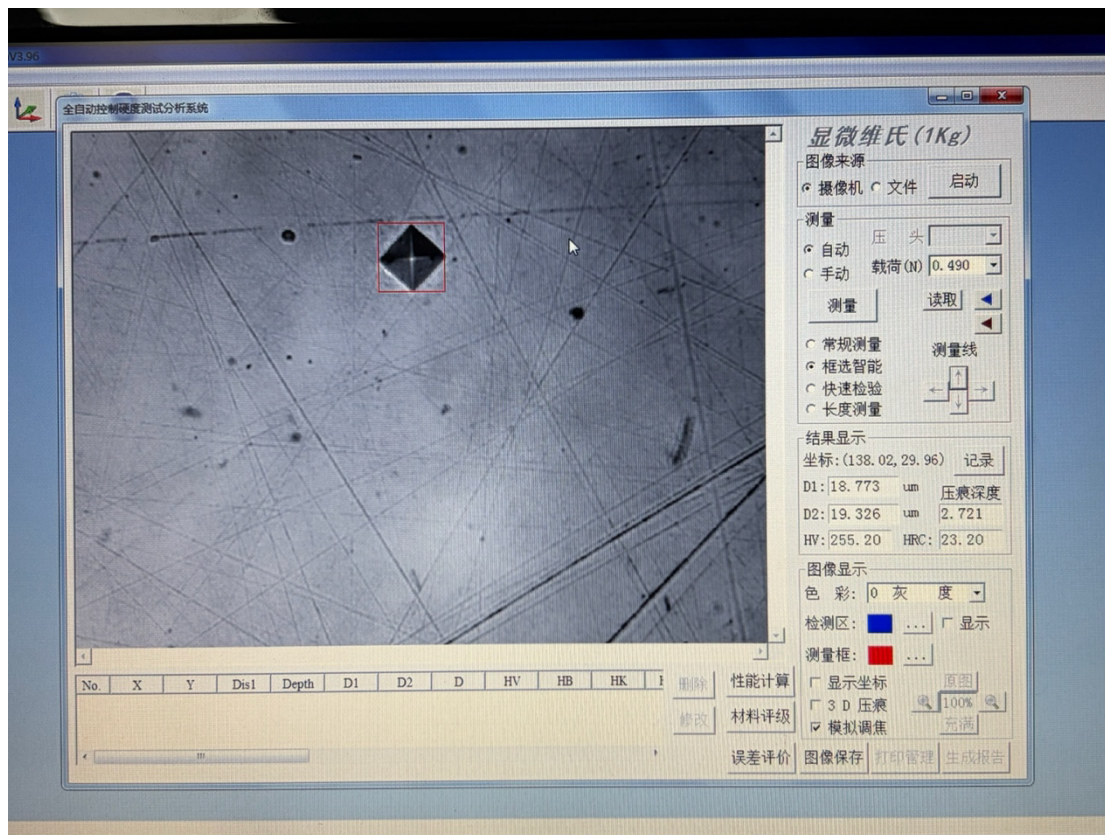


Figure 11. Schematic diagram of the Vickers hardness test

Vickers hardness testing was carried out using a Vickers hardness tester. The measured value was 255.20 HV, while the corresponding standard value was 258 HV. The deviation is only 2.80 HV, indicating good agreement with the standard value. The possible causes of this slight difference include:

Indentation measurement error: Vickers hardness relies on image analysis to measure the diagonal lengths of the indentation. There may be extremely small discrepancies in the system's image-capturing accuracy or in the edge-detection algorithm.

Microstructural variation: Vickers hardness testing probes a very small region. Even slight local non-uniformities in grain size or precipitate distribution at the test point can lead to minor fluctuations in the measured hardness.

Stability of load application: Very small fluctuations in the applied test load during the indentation process can also slightly affect the final indentation size and thus the hardness value.

5. Discussions

5.1 Principles and applications of Brinell, Rockwell and Microhardness

Brinell hardness test

In this method, a hardened steel ball or carbide ball with a standardized diameter is used as the indenter and is pressed into the specimen surface under a constant test load. After maintaining the load for a specified time, the indenter is removed, and the Brinell hardness value is obtained by measuring the average diameter of the residual indentation on the specimen surface and calculating it using the standard formula or consulting a conversion table. A key feature of this method is the relatively large indentation area, which allows it to reflect the average properties of a comparatively large material volume. It is therefore particularly suitable for materials with coarse or non-uniform grain structures, such as castings, forgings, and annealed nonferrous metals. To ensure accuracy and comparability of the results, the ratio of the test force (F) to the square of the indenter diameter (D^2), i.e., F/D^2 , must be selected according to the material, thickness, and expected hardness range of the specimen, in accordance with the relevant standards.

Rockwell hardness test

This is a hardness testing method based on the measurement of indentation depth. The indenter can be either a diamond cone or a steel ball of specified diameter. During testing, a minor load is first applied (pre-load), followed by the application of a major load (total load). After maintaining the total load for the specified time, the major load is removed while the minor load is retained; the Rockwell hardness value is then determined from the increase in indentation depth. Different combinations of indenter type and test load correspond to different hardness scales, such as HRA, HRB, and HRC. Owing to its operational simplicity, rapid readout, and small indentation size, this method is widely used for quality control of finished and semi-finished products in production environments, and is applicable to a broad range of materials from relatively soft to very hard.

Microhardness test

This method is essentially a Vickers hardness test conducted under small loads. It uses a diamond pyramid indenter with an included angle of 136° between opposite faces. Under a relatively low test force, the indenter is pressed into the material surface. After unloading, the average length of the two diagonals of the indentation is precisely measured using an optical system, and the hardness value is calculated accordingly. Because the applied load is very small, the resulting indentation is extremely tiny. This makes the method particularly suitable for evaluating the properties of thin layers, coatings, small components, specific phase constituents, or local regions within a material.

5.2 Which two hardness values of the same sample are close to each other? and Why?

For the same metallic specimen, the Brinell hardness (HB) value is usually close to the Vickers/microhardness (HV) value. This is mainly because the calculation principles of these two methods are essentially the same: both define hardness based on the average pressure over the indentation surface area. Brinell hardness is calculated using the curved surface area of a spherical indentation, whereas Vickers hardness is calculated using the inclined surface area of a pyramidal indentation. Despite the different indenter geometries, their physical definitions are similar. In addition, under conditions of geometric similarity, the two methods respond in a consistent way to plastic deformation behavior, thus yielding numerically comparable results.

By contrast, Rockwell hardness (HR) is a relative value derived from a specific depth-measurement scale. Its definition and dimensional nature are fundamentally different from those of Brinell hardness (HB) and Vickers hardness (HV). Therefore, these values cannot be directly compared and can only be approximately converted using empirical conversion tables.

5.3 What is the principle of indentation geometric similarity?

The principle of indentation geometric similarity is the fundamental theoretical basis for ensuring the comparability and validity of hardness test results. This principle requires that, for the same material, the indentations produced under different combinations of test force and indenter size must maintain similar geometric shapes.

In the Brinell hardness test, application of this principle is crucial. To preserve geometric similarity of the indentation, the angle ϕ (i.e., the central angle corresponding to the indentation diameter d and the indenter diameter D) must remain constant. From the relevant formulas, maintaining a constant ϕ requires that the ratio of the test force F to the square of the indenter diameter D^2 (i.e., F/D^2) be kept constant. Table 1 in the experimental guidelines specifies standard F/D^2 values that should be selected according to different material types and hardness ranges. In addition, the standards require that the measured indentation diameter d lie within the range of $0.25D$ to $0.60D$. If the value falls outside this range, it indicates that geometric similarity of the indentation has not been effectively achieved, rendering the test result invalid and necessitating retesting under adjusted conditions.

By adhering to the principle of geometric similarity, consistent and comparable hardness values can be obtained for the same material using different equipment specifications and in different laboratories, provided that an appropriate F/D^2 ratio is selected. This enables the standardization of hardness measurements.

6. Conclusions

This experiment systematically carried out metallographic microstructure observation and hardness testing, and successfully achieved the intended teaching and experimental objectives. In terms of procedure, standard metallographic specimens of GH4145 and IMI834 were first examined under an optical microscope. A stepwise magnification scheme from low to high magnification was adopted, and the aperture diaphragm and light intensity were adjusted to optimize image contrast and clarity, so that representative microstructures could be captured and recorded. Subsequently, Rockwell and Vickers hardness testers were used to measure the hardness of the

corresponding samples. The applied load, loading/holding time, and reading procedures were all strictly controlled in accordance with the operating standards. For each specimen, three different measurement locations were selected and averaged in order to improve the repeatability and statistical representativeness of the test data and thus ensure the reliability of the experimental results.

The results indicate that heat treatment has a significant regulatory effect on the microstructure and properties of the materials. Metallographic analysis shows that GH4145, after solution treatment at 1150 °C, develops a fully recrystallized equiaxed grain structure with a grain size of approximately 10–15 μm , accompanied by fine and uniformly distributed precipitates. This suggests that this heat treatment condition can produce a microstructure with good uniformity and stable performance. For the IMI834 titanium alloy, three typical microstructural morphologies were successfully obtained: equiaxed grain structure, bimodal grain structure, and a needle-like structure with characteristic “Widmanstätten-type” features. The differences among these three microstructures in terms of α phase morphology, remnants of prior β grain boundaries, and phase distribution directly correspond to different heat treatment routes, and clearly reveal the way in which heat treatment controls phase constitution and grain structure.

The hardness test data exhibit high consistency and credibility. The measured Rockwell hardness is 47.0 HRC, compared with a standard value of 47.1 HRC, giving a deviation of only 0.1 HRC; the Vickers hardness result is 255.2 HV, compared with a standard value of 258 HV, giving a deviation of 2.8 HV. These deviations lie within the normal fluctuation range of the instruments and experimental operations, indicating a high degree of standardization in the hardness testing procedure and good accuracy of the data. Overall, this experiment has clearly demonstrated, in a qualitative sense, the microstructural evolution of GH4145 and IMI834 under different heat treatment conditions, and, in a quantitative sense, has validated the reliability of the testing methods through hardness data that are close to the standard values. This provides a solid basis for further establishing and understanding the quantitative relationships among “process–microstructure–property”, and shows that the experiment has successfully fulfilled its intended objectives.